

Bidirectional Mass Spectrometry Modeling Aided by Auxiliary Intermediate Steps From Graph Transformation Tools

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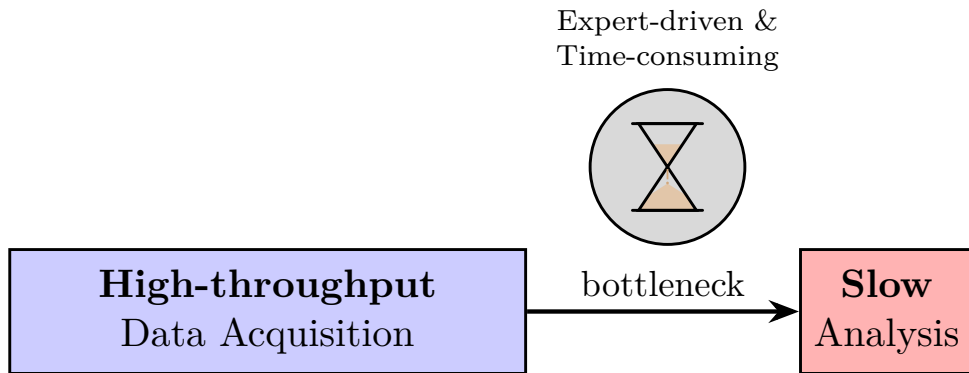
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2026-08-27

Outline

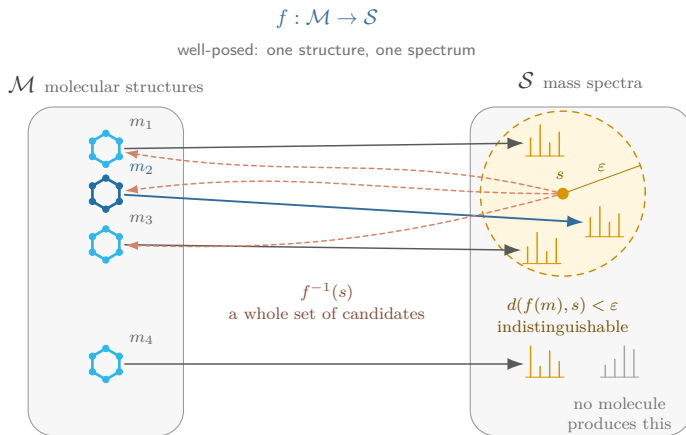
- 1 Problem
- 2 Background
- 3 Data
- 4 ML-Modeling
- 5 Results
- 6 Conclusions

Motivation



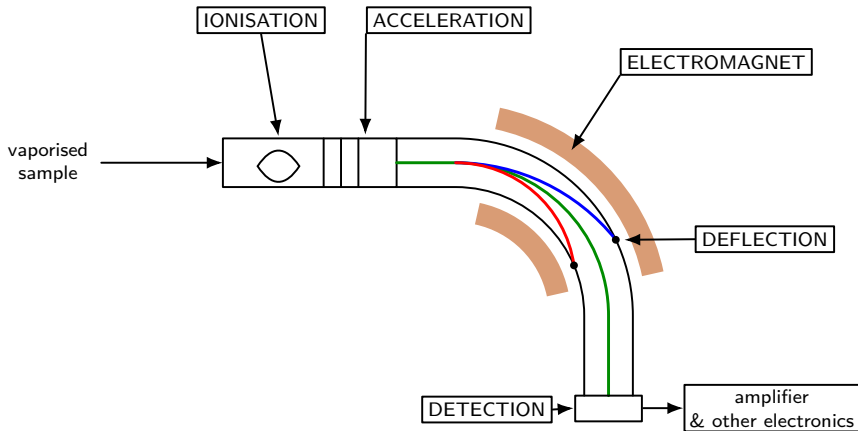
- Mass spectrometry is widely used, but analysis is the challenge
- Data acquisition is fast, interpretation slow $\Rightarrow$ the bottleneck

Problem Statement



Structure elucidation is the inverse of a map that is not invertible.

Mass Spectrometer

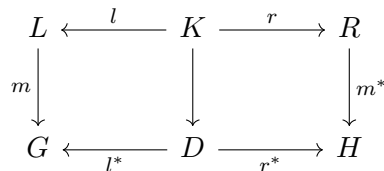


Modelling Mass Spectra – An Open Problem

- **Rule-based** but limited to rule set
- **Data-driven** models do not generalize
- **Ab initio** does not scale
- currently <20% of detected compounds are identified

MØD – A Graph Transformation Tool

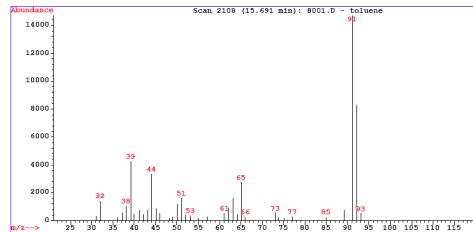
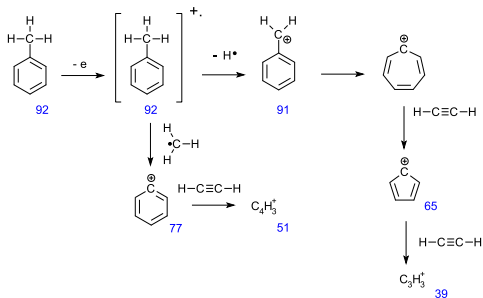
- Molecules as labelled graphs, reactions as *rewrite rules*
- Repeated application builds the *derivation graph* of a reaction network
- A rule is a span $L \leftarrow K \rightarrow R$:
 - L – what must be present
 - K – what is preserved (context)
 - R – what it becomes



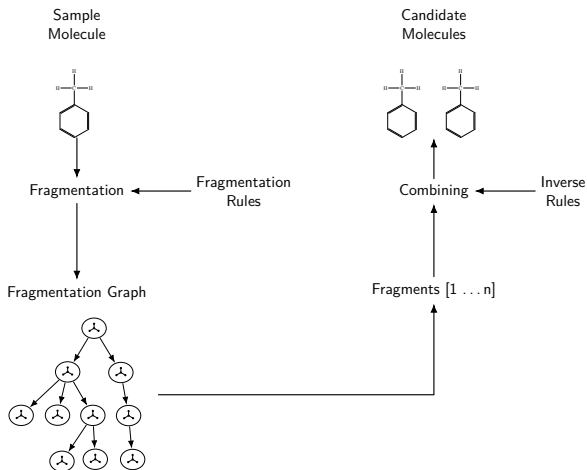
Host graph $G \rightarrow$ product graph H

Data Sources

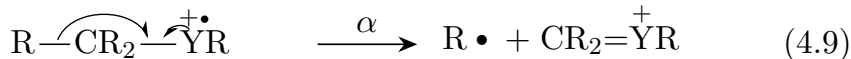
- Mass spectrometry data from NIST → SMILES strings & spectra
- Curated EI fragmentation reactions → encoded Graph Transformations



Data Generation Overview



Sample Fragmentation Rules

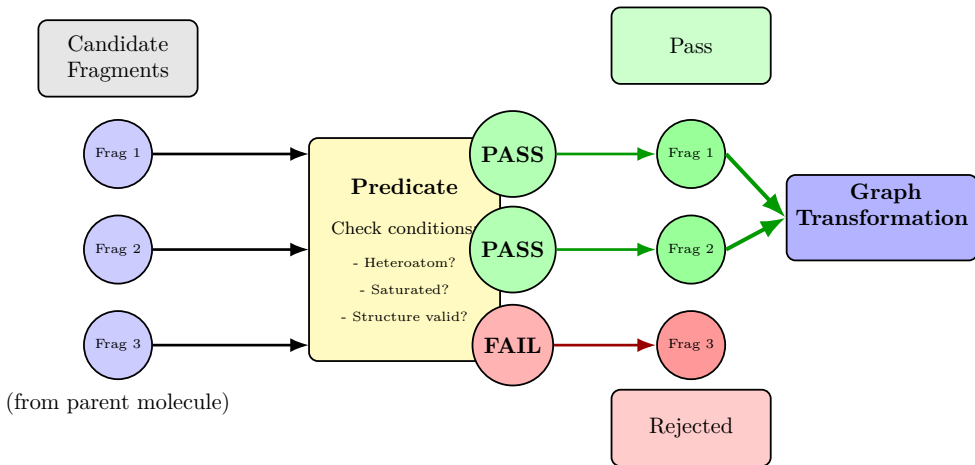


```

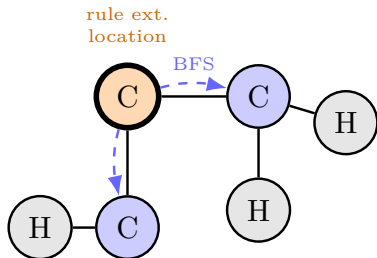
1      IMS_4_9 = mod.Rule.fromDFS(
2          s =
3              "[C]1[C]2([C]3)([C]4)[_A+.]5"
4              ">>"
5              "[C.]1" " ." "[C]2([C]3)([C]4){=}[_A+]5",
6          name =
7              "radical induced (alpha-)cleavage for a saturated site"
8              " 4.9"
9              " SR1R3R4Y5"
10     )

```

Predicate

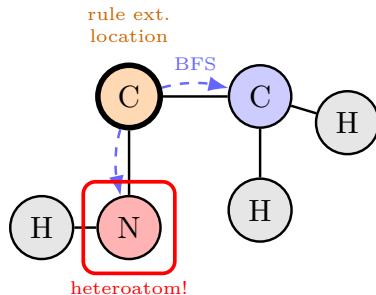


Predicate – Alkyl Group



VALID: Alkyl Group

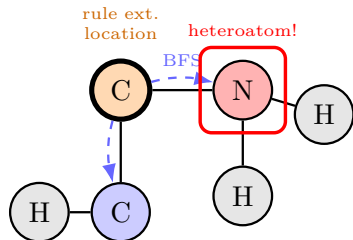
BFS neighbors $\subseteq \{H, C\} \checkmark$



INVALID: Not Alkyl

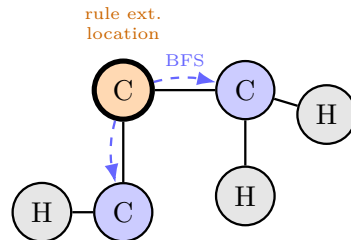
BFS neighbors $\not\subseteq \{H, C\} \times$

Predicate – Heteroatom Group



VALID: Heteroatom Group

$$\text{BFS} \setminus \{\text{H}, \text{C}\} \neq \emptyset \checkmark$$

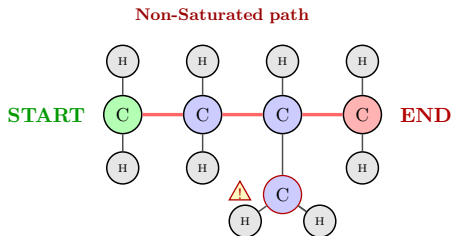
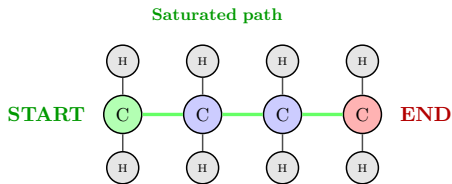


INVALID: No Heteroatom

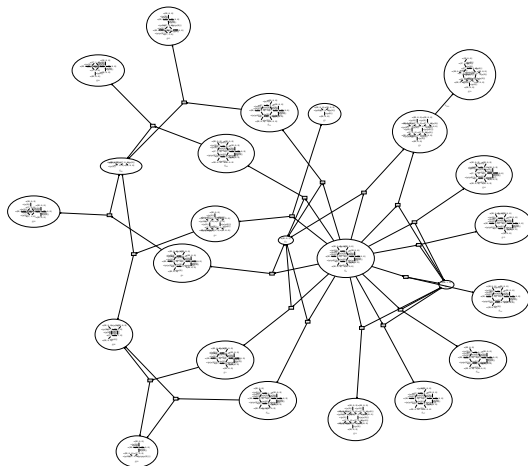
$$\text{BFS} \setminus \{\text{H}, \text{C}\} = \emptyset \times$$

Predicate – Saturation

Goal: Find saturated path
Method:
Search from **start** to **end**
following C-C single bonds



Sample Fragmentation Graph



Bidirectional Framework

$\mathcal{M}$ molecular structures



molecule graph

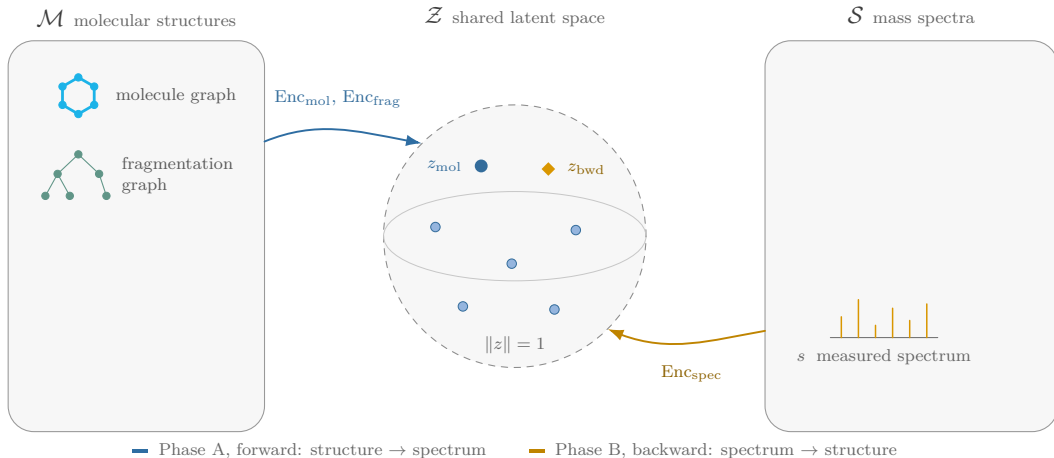


fragmentation
graph

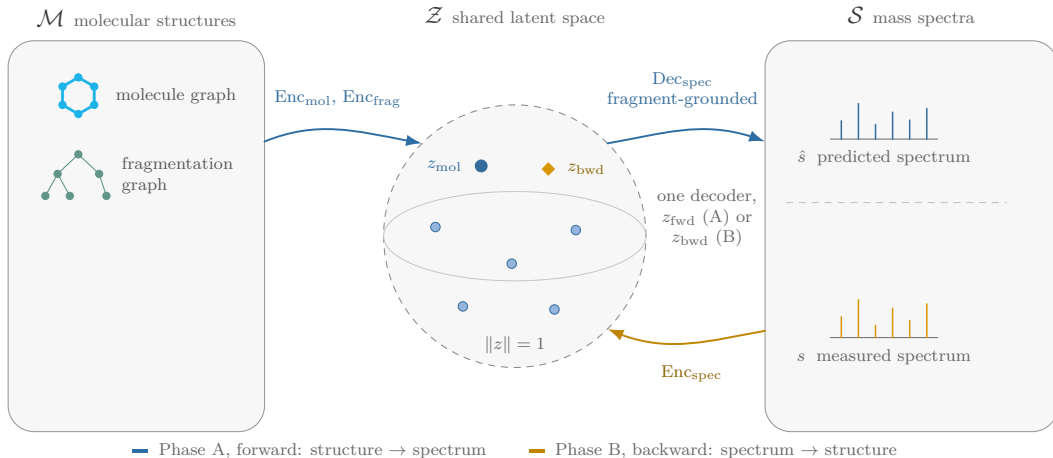
$\mathcal{S}$ mass spectra



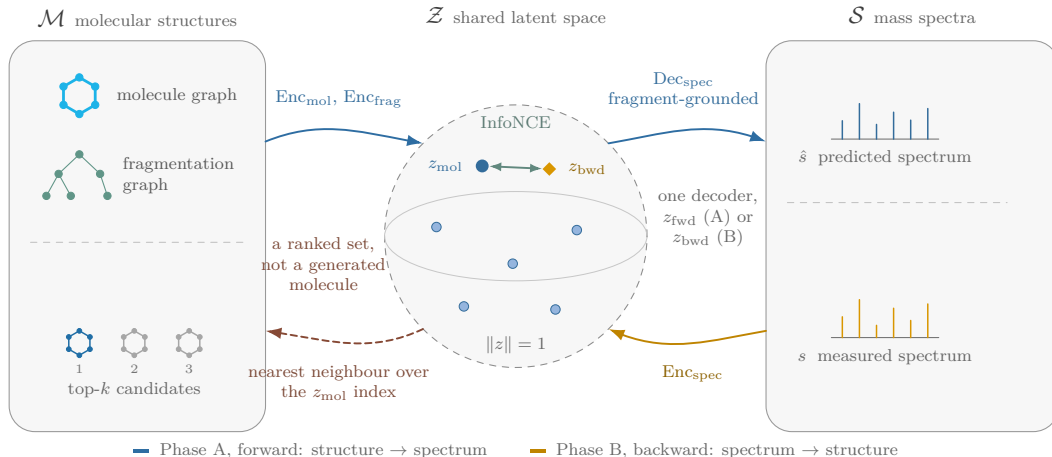
Bidirectional Framework



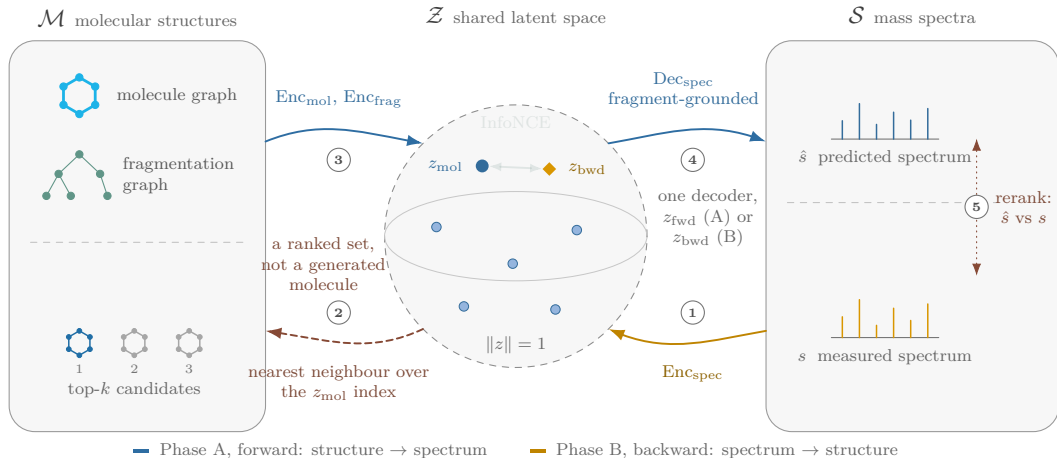
Bidirectional Framework



Bidirectional Framework



Bidirectional Framework – at Query Time



Results – Ablation

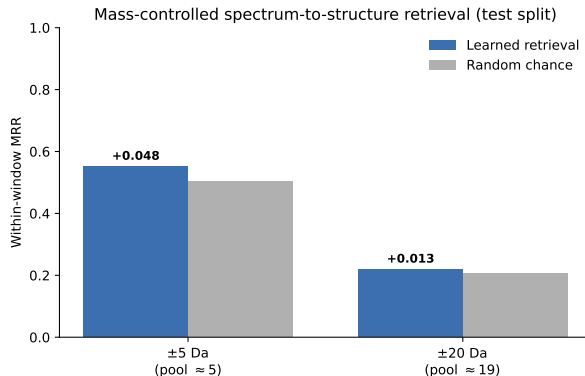
Mass Sets the Bar

Held-out test split · $N = 13$ queries · gallery of 66 EI compounds

Ranking signal	MRR
True parent mass (oracle)	0.923
Spectrum-implied parent mass	0.462
Mass mask + cosine	0.498
Learned embedding	0.112

⇒ uncontrolled retrieval scores measure **mass agreement**, not chemistry

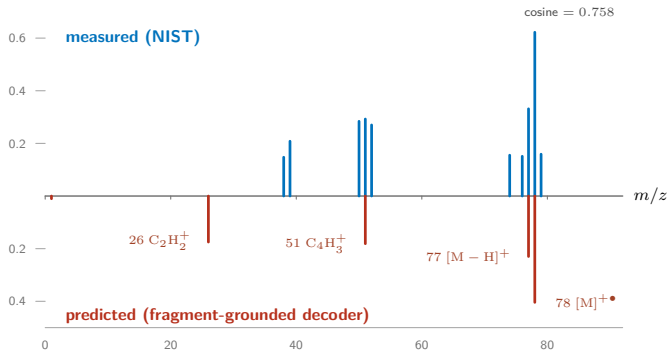
Results – Mass-Controlled Retrieval



±5 Da: 0.551 vs. chance 0.503 ($\Delta +0.048$, mean pool 5.4)
±20 Da: 0.219 vs. chance 0.206 ($\Delta +0.013$, mean pool 18.8)

Inside a window mass is $\approx$ constant $\Rightarrow$ the margin is structure the mass cannot explain.

Results – The Forward Model Is Molecule-Specific



Conclusions

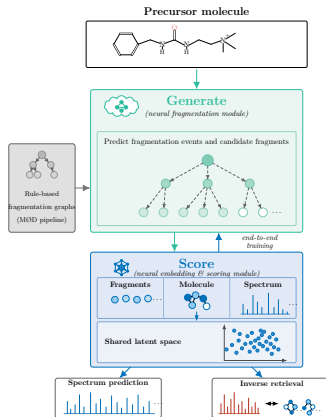
Contributions

- **Fragment-grounded forward model:** intensities scattered from the molecule's own derived fragments $\Rightarrow$ molecule-specific *by construction*
- **Mass-controlled evaluation:** separates learned structure from parent-mass filtering

Supporting work: bidirectional framework ($f : \mathcal{M} \rightarrow \mathcal{S}$, $f^{-1} : \mathcal{S} \rightarrow \mathcal{M}$) + rule-extension DSL

Outlook & Literature comparison

Learn to *Generate* Fragmentation



References

- [1] Adamo Young, Hannes Röst, and Bo Wang. “Tandem mass spectrum prediction for small molecules using graph transformers”. In: *Nature Machine Intelligence* 6.4 (Apr. 2024), pp. 404–416. ISSN: 2522-5839. DOI: 10.1038/s42256-024-00816-8.
- [2] Yannek Nowatzky et al. “FIORA: Local neighborhood-based prediction of compound mass spectra from single fragmentation events”. In: *Nature Communications* 16.1 (Mar. 2025). ISSN: 2041-1723. DOI: 10.1038/s41467-025-57422-4.
- [3] Mehdi B. Hamaneh et al. “Systematic Assessment of Deep Learning-Based Predictors of Fragmentation Intensity Profiles”. In: *Journal of Proteome Research* 23.6 (May 2024), pp. 1983–1999. ISSN: 1535-3907. DOI: 10.1021/acs.jproteome.3c00857.
- [4] Ivana Blaženović et al. “Software Tools and Approaches for Compound Identification of LC-MS/MS Data in Metabolomics”. In: *Metabolites* 8.2 (May 2018), p. 31. ISSN: 2218-1989. DOI: 10.3390/metabo8020031.
- [5] Jakob L. Andersen et al. “An intermediate level of abstraction for computational systems chemistry”. In: *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 375.2109 (Nov. 2017), p. 20160354. ISSN: 1471-2962. DOI: 10.1098/rsta.2016.0354.
- [6] MØD Developers. *Derivation Graph Strategy (dgStrat) — MØD Documentation*. Accessed 2026-03-24. 2022. URL: <https://imada.sdu.dk/u/daniel/DM840-2022/Material/mod-doc/dgStrat/index.html> (visited on 03/24/2026).
- [7] V8rik. *Toluene Fragmentation*. 2008. URL: <https://upload.wikimedia.org/wikipedia/commons/6/6c/TolueneFragmentation.svg> (visited on 01/20/2026).
- [8] Paginazero, Ronhjones. *Toluene Spectra*. 2017. URL: https://upload.wikimedia.org/wikipedia/commons/2/2c/Spettro_di_massa_del_toluene.PNG (visited on 01/20/2026).
- [9] Fred W McLafferty and František Tureček. *Interpretation von Massenspektren*. ger. Spektrum Lehrbuch. Berlin [u.a.]: Spektrum Akad. Verl., 1995. ISBN: 9783642398483.
- [10] Samuel Goldman, Janet Li, and Connor W. Coley. “Generating Molecular Fragmentation Graphs with Autoregressive Neural Networks”. In: *Analytical Chemistry* 96.8 (Feb. 2024), pp. 3419–3428. ISSN: 1520-6882. DOI: 10.1021/acs.analchem.3c04654.

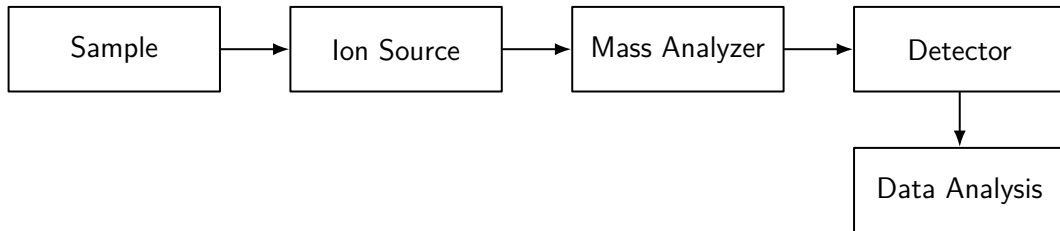
Discussion

Questions?

Supplementary Figures

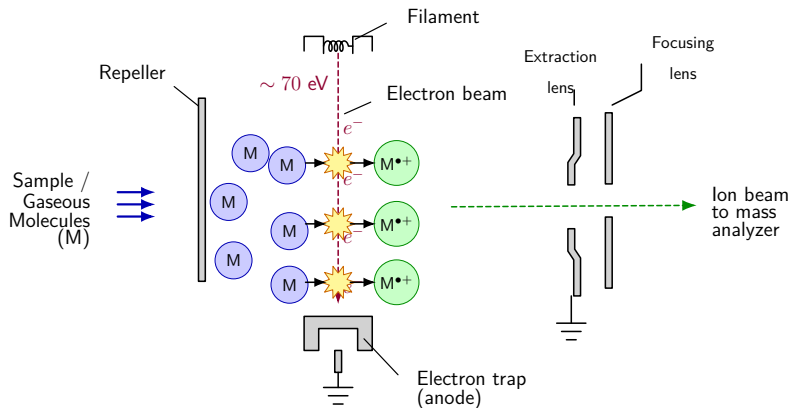
- Instrumentation & ionization methods
- Molecular representation
- Model architecture — training & inference
- The residual MLP block
- Latent-space alignment (InfoNCE)
- Decoder conditioning
- Spectral comparison measures
- Data-generation cost
- What the speedup enabled
- Train / validation / test split

Mass Spectrometer – Block Diagram



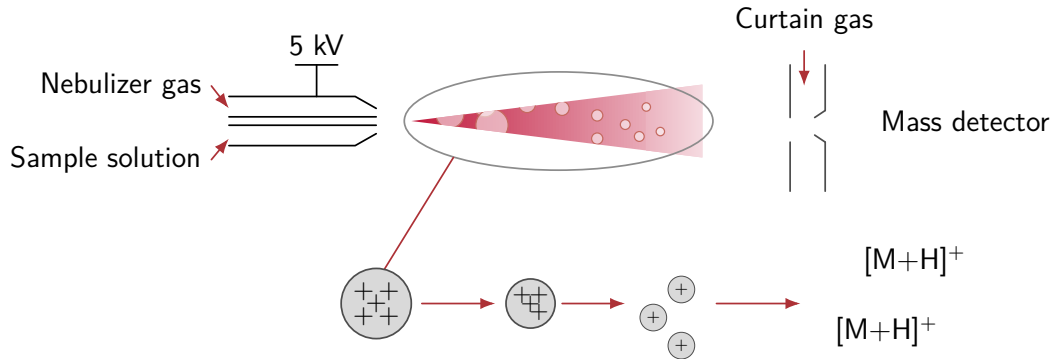
Inlet → ion source → mass analyzer → detector → data analysis.

Electron Ionization (EI)



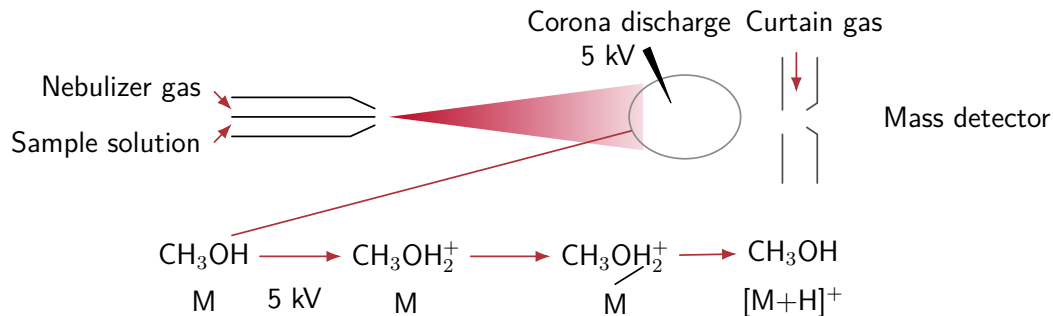
$\sim 70 \text{ eV}$ electrons eject one electron $\Rightarrow$ radical cation $[M]^{\bullet+}$, which then fragments.

Electrospray Ionization (ESI)



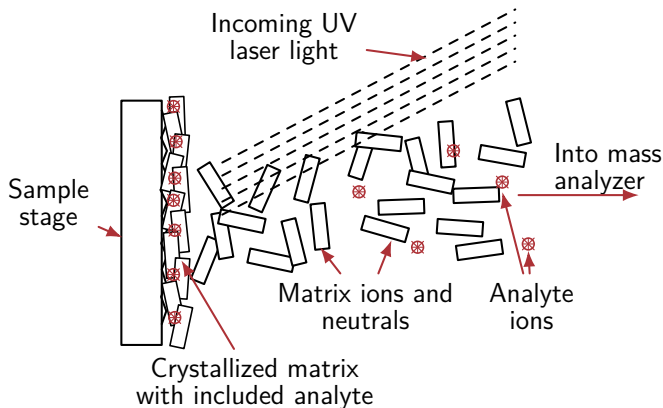
Soft ionization: charged droplets $\rightarrow$ gas-phase ions, typically $[M + H]^+$.

Atmospheric Pressure Chemical Ionization (APCI)



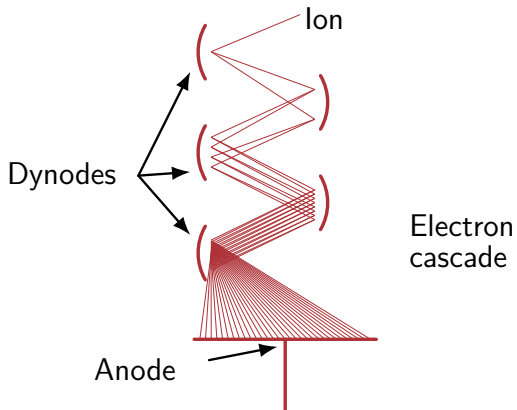
Corona discharge makes reagent ions; proton transfer gives $[\text{M} + \text{H}]^+$.

Matrix-Assisted Laser Desorption/Ionization (MALDI)



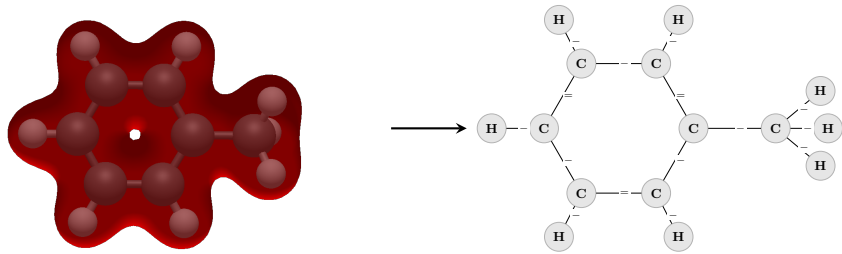
Laser desorption from a matrix; predominantly singly charged ions, suited to large biomolecules.

Electron Multiplier Detector



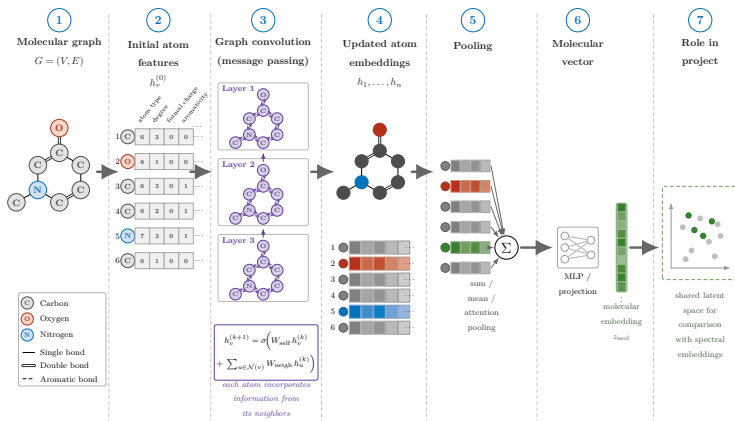
Cascading secondary electrons across dynodes amplify a single ion into a measurable current pulse.

Molecule as a Graph



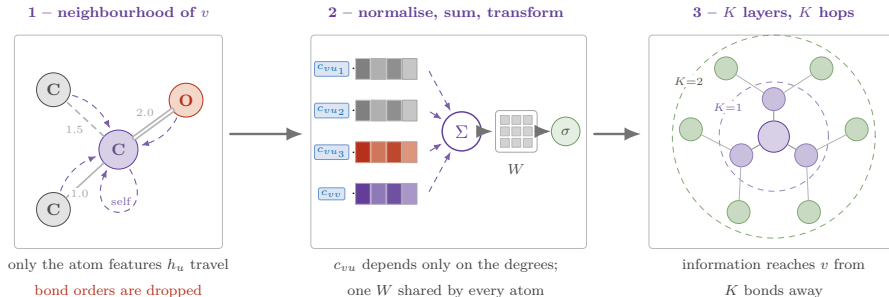
Geometry and electron density are abstracted away; connectivity is kept.

Graph-Convolutional Molecular Encoding



Atom features $\rightarrow$ message passing $\rightarrow$ pooling $\rightarrow z_{mol}$ in the shared latent space.

What a Graph Convolution Does

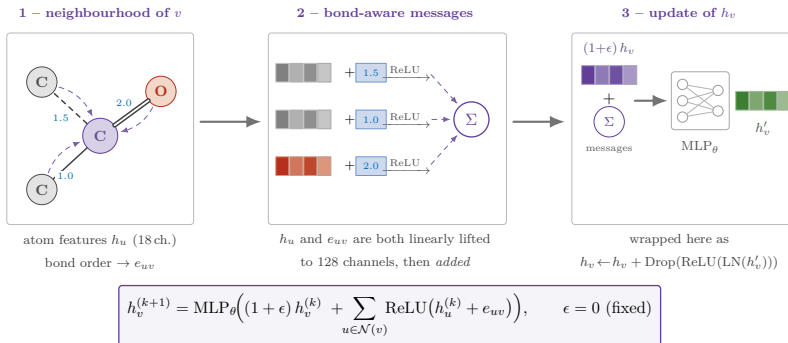


$$h_v^{(k+1)} = \sigma \left(W^{(k)} \sum_{u \in \mathcal{N}(v) \cup \{v\}} c_{vu} h_u^{(k)} \right), \quad c_{vu} = \frac{1}{\sqrt{\hat{d}_v \hat{d}_u}}$$

no e_{uv} in the sum $\Rightarrow$ single, double and aromatic bonds send the same message

Each atom is replaced by a degree-weighted average of its neighbours; K layers $\Rightarrow$ K -hop context.

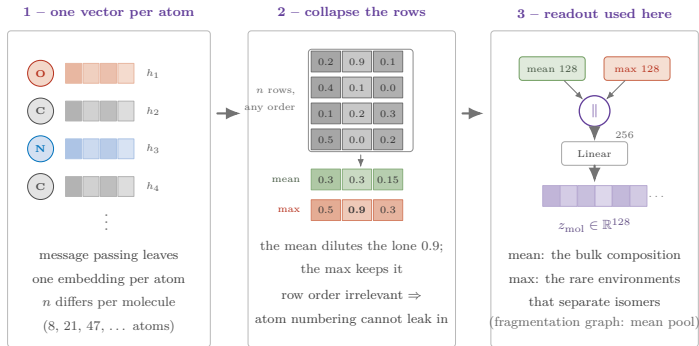
Why GINEConv – Bond Orders Enter the Message



A GCN averages neighbours and *drops* the bond, so what survives is composition — i.e. mass. GINE adds the bond embedding to every message and *sums* (1-WL expressive) $\Rightarrow$ constitutional isomers stay distinguishable.

Here: 3×128 layers, LayerNorm + residual, mean $\parallel$ max readout $\rightarrow z_{\text{mol}} \in \mathbb{R}^{256}$.

Pooling – n Atom Vectors $\rightarrow$ One Molecule Vector



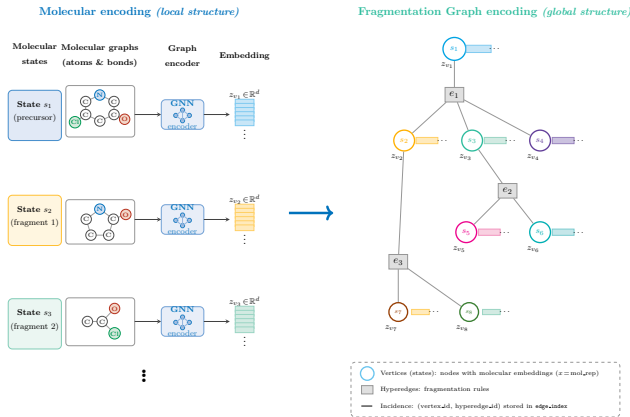
$$z_{\text{mol}} = W \left[\text{mean}_{v \in V} h_v \parallel \max_{v \in V} h_v \right] \in \mathbb{R}^{128}, \quad \text{unchanged by any permutation of } V$$

Two-Level Representation – Overview



Local graph encoders for atom/bond chemistry, global hypergraph encoders for multi-step fragmentation.

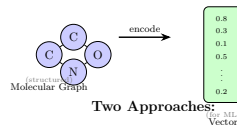
Two-Level Molecular Latent Representation



Stage 1: state $\rightarrow$ embedding. Stage 2: embeddings become vertex features of the fragmentation hypergraph.

The Graph Representation Challenge

Graph Representation Challenge



Two Approaches:

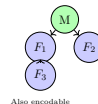


- ✓ Better performance
- ✗ Not invertible



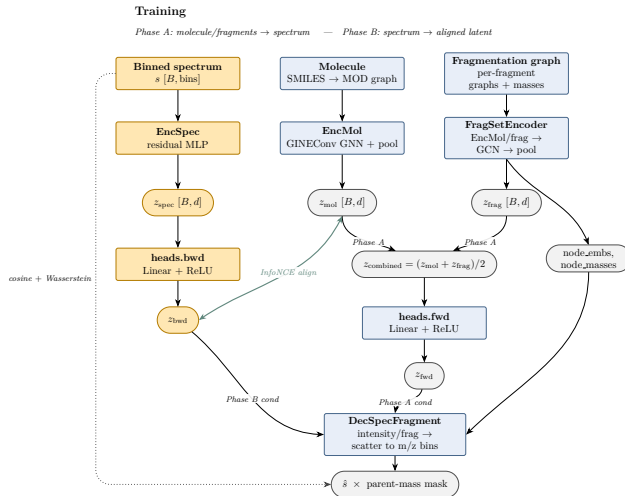
- ✓ Structured info
- ✓ For fragments too

Fragmentation Trees



Fingerprints score well but are not (easily) invertible — which matters for the inverse direction.

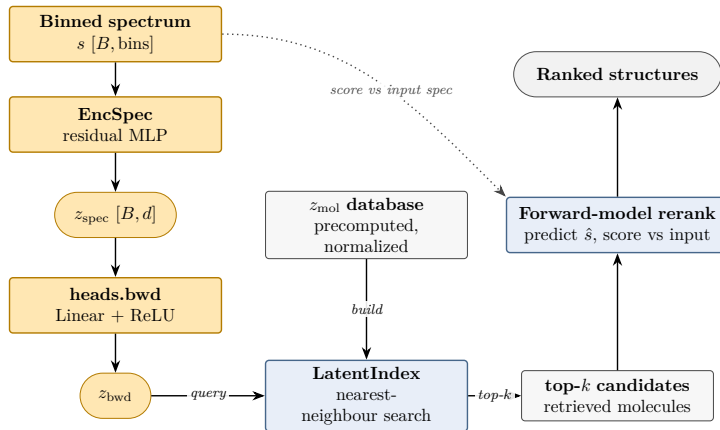
Architecture – Training



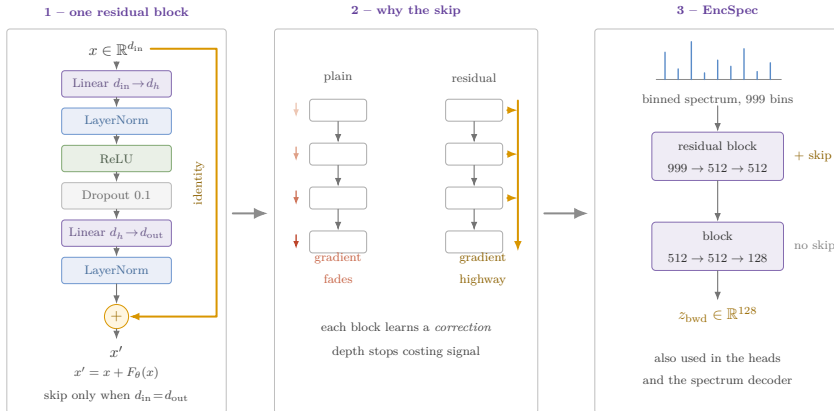
Architecture – Inference

Inference / retrieval

Query with z_{bwd} into the molecule-latent index, then rerank with the forward model

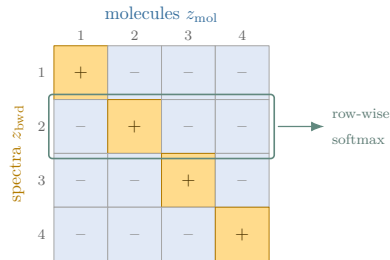
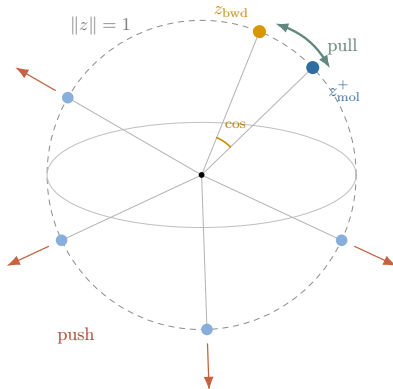


The Residual MLP Block



$x' = x + F_{\theta}(x)$ — the block learns a *correction*, not the whole map.

Phase B – InfoNCE Alignment



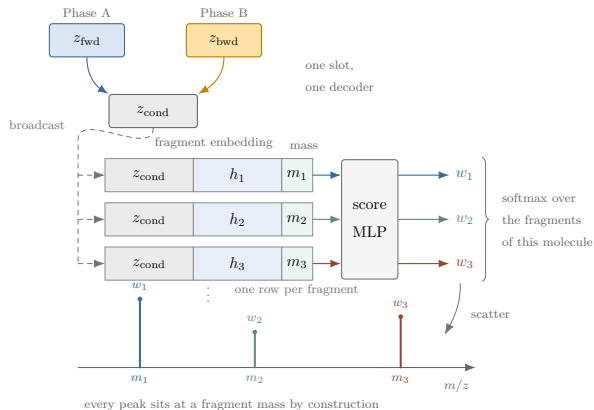
$$\mathcal{L}_{\text{NCE}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(z_{\text{bwd}}^i \cdot z_{\text{mol}}^i / T)}{\sum_{j=1}^B \exp(z_{\text{bwd}}^i \cdot z_{\text{mol}}^j / T)}$$

$T = 0.07$ batches sorted by parent mass

$\Rightarrow$ the negatives are near-isobaric

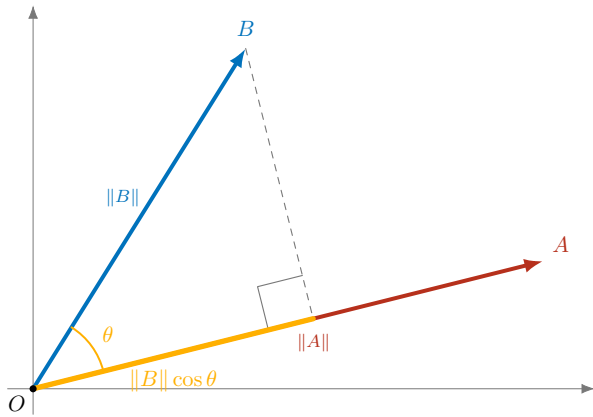
One shared space · diagonal = positive, rest of the batch = negatives

Conditioning the Decoder



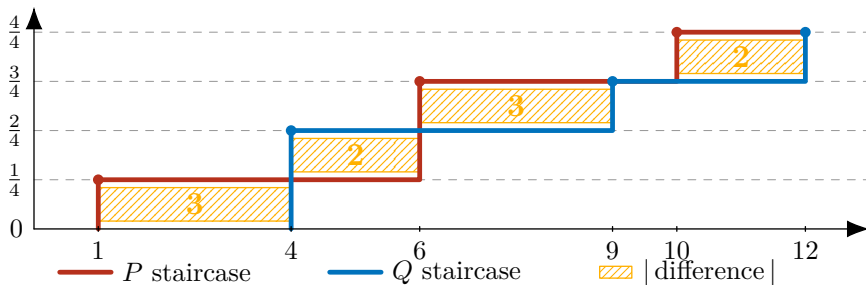
One decoder, one slot: Phase A puts z_{fwd} in it, Phase B z_{bwd} .

Cosine Similarity – Geometry



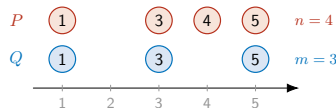
$\cos \theta = (A \cdot B) / (\|A\| \|B\|)$ — direction only, hence scale-invariant.

Wasserstein Distance – Illustration



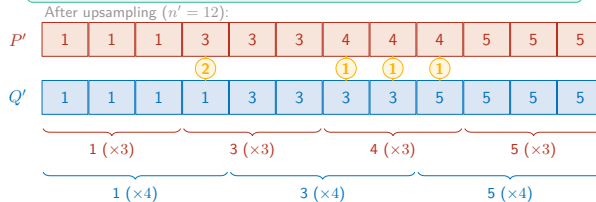
Sorted vectors as cumulative staircases; the distance is the shaded transport area, normalized by n .

Wasserstein Distance – Upsampling



$\text{LCM}(4, 3) = 12$ — upsample to equal length

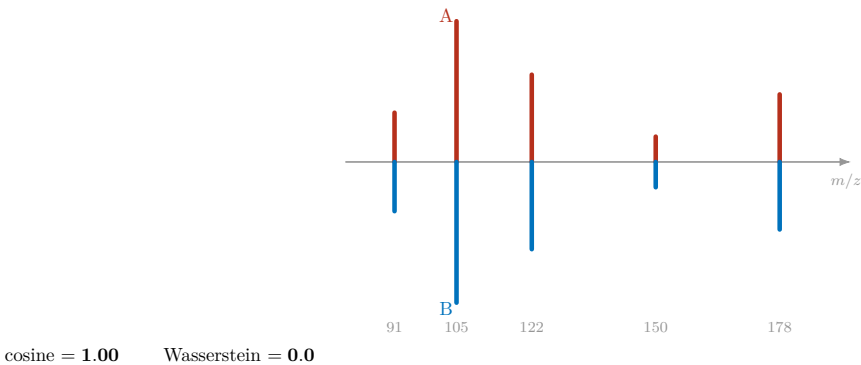
repeat each P element $\times 3$ · repeat each Q element $\times 4$



Unequal peak counts are brought to a common length $\text{LCM}(n, m)$ by repetition, preserving each value's share.

Cosine vs. Wasserstein – (a) Aligned Spectra

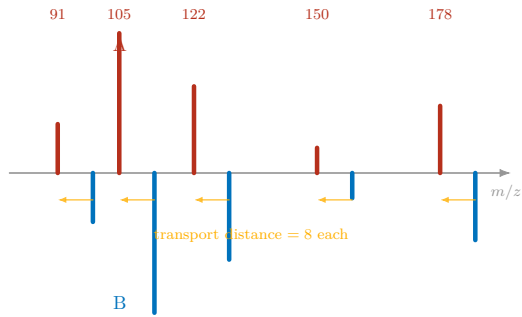
(a) Aligned spectra — identical, $\theta = 0$



Every B peak meets its A counterpart: cosine = 1 ($\theta = 0$), Wasserstein = 0 (no mass to move).

Cosine vs. Wasserstein – (b) B Shifted +8 Da

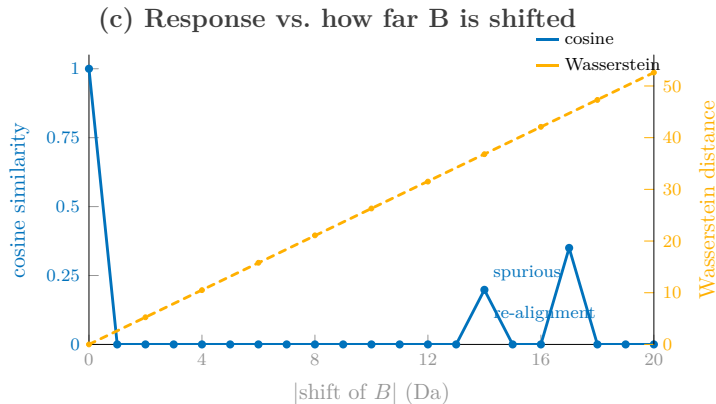
(b) B shifted +8 Da — peaks miss, $\theta \rightarrow 90^\circ$



cosine = **0.00** (no shared peaks) Wasserstein = **21.1**

Cosine counts only *shared* positions $\Rightarrow$ collapses to 0; Wasserstein transports the mass back and stays finite (21.1).

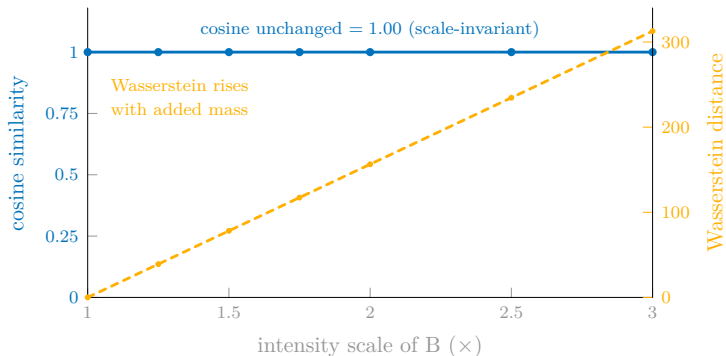
Cosine vs. Wasserstein – (c) Response vs. Shift



Cosine drops to zero at once, with spurious spikes at ~ 14 and 17 Da; Wasserstein rises linearly with displacement.

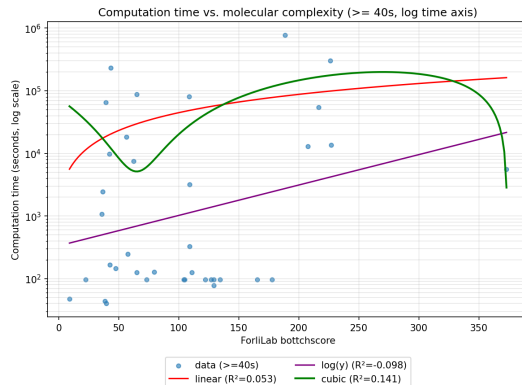
Cosine vs. Wasserstein – (d) Response vs. Scaling

(d) Response vs. B's peak intensity (positions aligned)



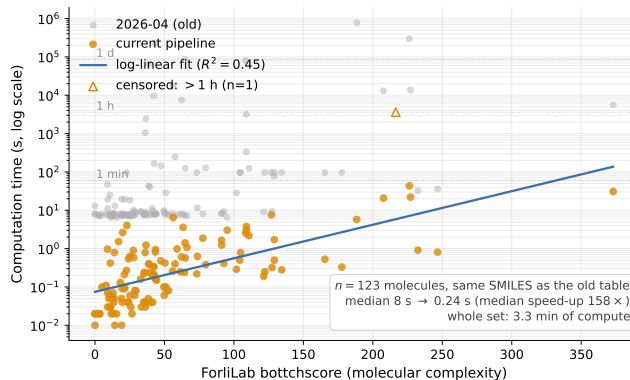
Cosine unchanged at 1.00 (direction, not magnitude); un-normalized Wasserstein grows with the added intensity.

Fragmentation Computation Times



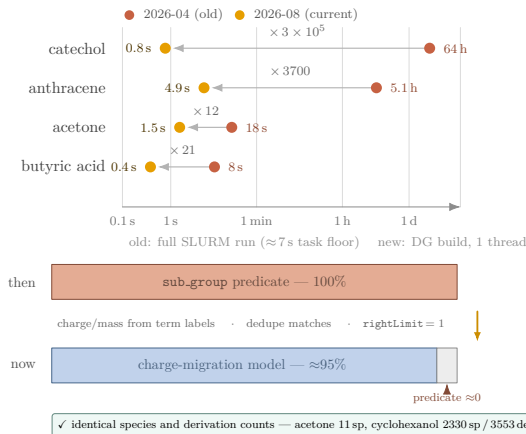
Time vs. molecular complexity (trivial expansions < 40 s excluded): positive trend, but huge variance at equal complexity.

Fragmentation Computation Times – Re-measured



Same molecules, current code · the trend survives, the level drops

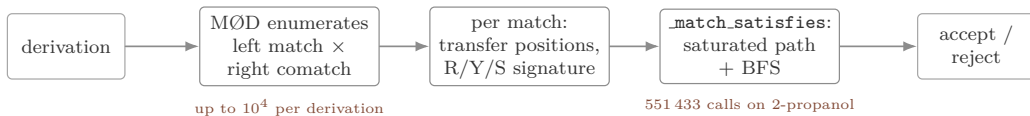
Fragmentation Cost – Then vs Now



Same fragments, more chemistry · the bottleneck moved

Fragmentation Cost – What Moved

The hot path, 2026



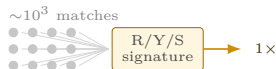
What moved it

① Read, don't rebuild

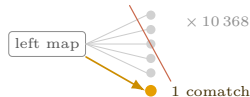
~~term → SMILES → molecule → q, m~~

term label → q, m

② Memoise the traversal



③ Cap the comatch fan



Where the Data-Generation Time Goes

MØD data generation — sampled call tree (folic_acid, forward-only, 1 thread)

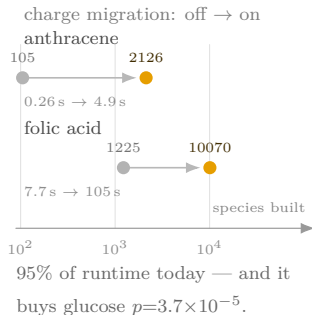
width = wall time · 11.662 stack samples over 117.9 s of build (imports pre-warmed) · 86% of samples are inside MØD's C++ matching loop with no Python frame below



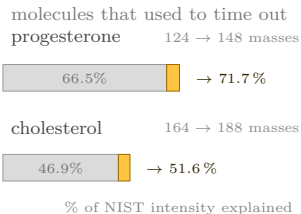
Sampled call tree (folic acid, forward-only, 1 thread): 86% of samples sit inside MØD's C++ matching loop.

What the Speedup Bought

① New chemistry



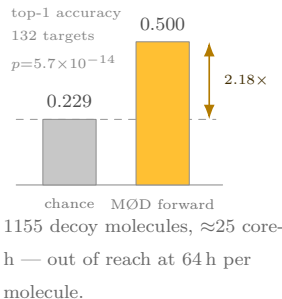
② The hard tail



✓ aldosterone: killed at 12 h30
→ 1 h14. A timeout writes no
dump.

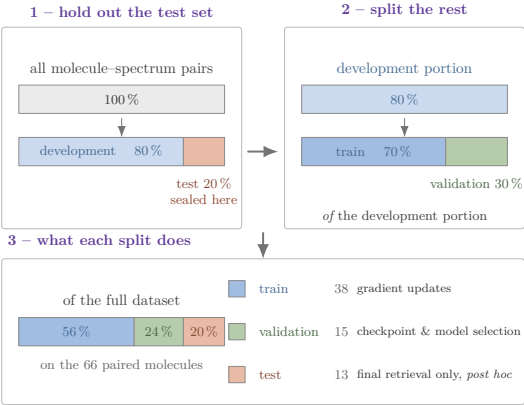
③ 1340 decoys

same-formula discrimination



Faster is not the point · what the compute was spent on is

Train / Validation / Test – Two-Stage Split



shuffle=False — deterministic, reproducible by construction

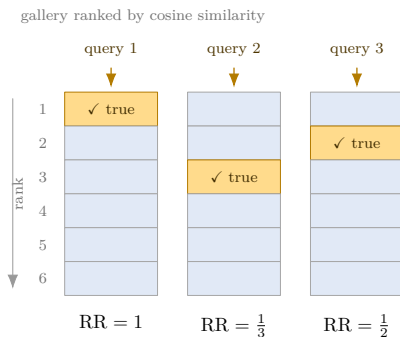
$$56 / 24 / 20 \% \Rightarrow 38 / 15 / 13 \text{ molecules}$$

Splits, Gallery and Provenance

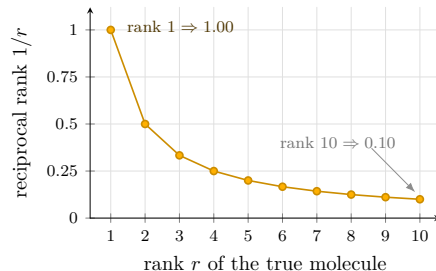
Molecules with paired dumps + spectra	66
train / validation / test	38 / 15 / 13
Retrieval gallery (train + val + test)	66
Mass-controlled queries (test)	13
Training runs behind the numbers	1
Spectrum bins (1–1000 Da, 1 Da)	999

All reported numbers come from one training run (`ml_checkpoint.pt`, 2026-07-06) — no seed averaging, no model selection on the test split.

Mean Reciprocal Rank (MRR)



$$\text{MRR} = \frac{1}{N} \sum_{i=1}^N \frac{1}{r_i} = \frac{1 + \frac{1}{3} + \frac{1}{2}}{3} = 0.61$$



Only the rank of the *true* molecule counts · the top of the list dominates

Analytic Chance Level in a Mass Window

A query restricted to $\pm W$ Da leaves a pool of m candidates, exactly one of which is correct. Ranking that pool at random puts the correct molecule at rank r with probability $1/m$, so

$$\text{MRR}_{\text{chance}}(m) = \mathbb{E}\left[\frac{1}{r}\right] = \frac{1}{m} \sum_{r=1}^m \frac{1}{r} = \frac{H_m}{m}, \quad \text{chance} = \frac{1}{N} \sum_{i=1}^N \frac{H_{m_i}}{m_i}$$

H_m : the m -th harmonic number; m_i : pool size of query i .

m	1	2	3	5	10	20
H_m/m	1.000	0.750	0.611	0.457	0.293	0.180

The average is taken *per query*, not at the mean pool size: H_m/m is convex, so plugging in $\bar{m} = 5.4$ would give ≈ 0.44 instead of 0.503 — the per-query bar is the **harder** one.

Honest assessment

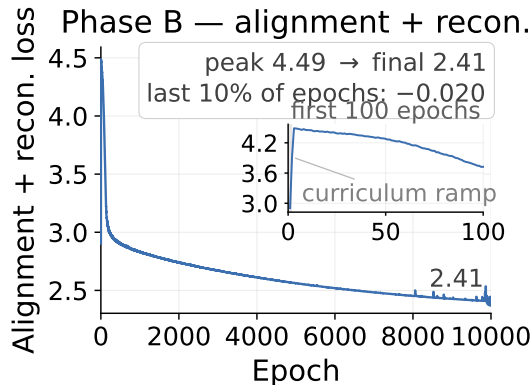
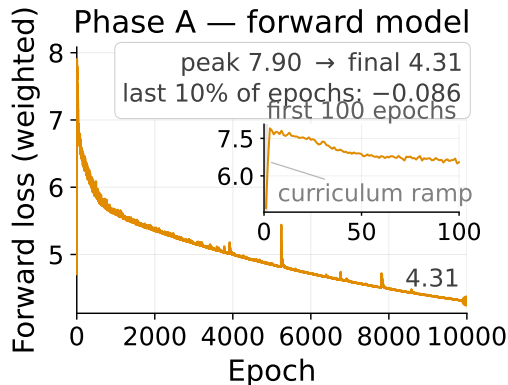
- Beyond-mass margin is *small*: +0.048 / +0.013 MRR on 13 queries, single run
- Uncontrolled metrics — reranking included — are mass filtering, not evidence
- Fragment branch buys molecule-specificity, not a retrieval win

Training Hyperparameters

Optimizer	Adam (one per phase)
Learning rate / weight decay	7.1×10^{-5} / 1.4×10^{-4}
Batch size	64
Epochs (Phase A + Phase B)	10 000 + 10 000
Shared latent dimension d	256
GINEConv layers / width	3 / 128
Fragmentation-graph GCN	2× GCNConv, mean pool
Dropout / normalization	0.43 / BatchNorm
Cosine / Wasserstein weight	1.59 / 1.17
Cycle (fwd / spec) weight	0.77 / 0.089
Align weight / temperature τ	0.30 / 0.07
Curriculum start / increment	0.73 / 0.17

Tuned entries: best of a 20-trial Optuna sweep; checkpoint selection = final epoch.

Training Loss – Both Phases



Training loss only · no per-epoch validation logged · checkpoint = final epoch

Model Size – Where the Parameters Are

Module	Parameters	Share
Molecule encoder ENCMOL ($3\times$ GINEConv, $w=128$)	168 320	9.9%
Fragmentation-graph GCN ($2\times$ GCNConv)	33 152	1.9%
Spectrum encoder ENCSPEC ($999\rightarrow 512\rightarrow 512\rightarrow 256$)	1 170 688	68.8%
Fragment-grounded decoder DECSPECFRAGMENT	198 657	11.7%
Task heads (fwd / bwd + loss $\log \sigma^2$)	131 586	7.7%
Total (trainable)	1 702 403	

The single 999×512 input matrix of the spectrum encoder is 30% of the whole model.